Data Science Project Report

Technical Analysis and Machine Learning for Swing-Trade Entry Timing: A Data Science Study of Oracle Corporation (ORCL), 1986–2026

Student(s) Name(s):

Santiago Cruz Lopez | **ID:** 2584023

7CS078/UZ1: Data Science

Online

Professor Khubaib Ahmed

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Wolverhampton, UK

The University of Wolverhampton

Abstract

Technical analysis assumes that past price activity – trend, momentum, and support/resistance structure – contains information that can be used to predict short-term price movement; an assumption that has long been challenged by the Efficient Market Hypothesis's assertion that prices already incorporate all available information (Fama, 1970).

This paper presents a data science pipeline to predict the price direction and range of Oracle Corporation (NYSE: ORCL) stock over a 21-trading-day period. Our approach combines technical analysis with statistical and machine learning algorithms, treating market efficiency as a null hypothesis to surpass. The pipeline uses daily OHLC price data from 1986 to 2026, covering 10,185 trading sessions. It incorporates indicators such as the Relative Strength Index (RSI), True Strength Index (TSI), multi-window Simple Moving Averages and Volume-Weighted Moving Averages, the Average True Range (ATR), and empirical support and resistance structures (Malkiel, 2003).

The architecture employs several machine learning techniques for a three-class classification of Buy, Hold, and Sell. It combines Logistic Regression, a feed-forward Neural Network, and Random Forest, along with Principal Component Analysis and K-Means for regime clustering, a Gaussian Hidden Markov Model for probabilistic regime identification, and Bidirectional LSTM for price forecasting.

Out-of-sample validation results show a modest but significant improvement over a naive baseline, with Random Forest achieving 40.5% balanced accuracy versus 33.3%. A hypothesis test on the Relative Strength Index (RSI) yielded a p-value of 0.001. However, the LSTM only outperformed a naive persistence baseline after being reformulated on stationary returns. This illustrates the value of transparent statistical models over opaque black-box approaches, reinforcing that a modest, well-validated edge is more credible than an inflated one.

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1. Introduction

Technical analysis is based on the premise that past price action, including trends, momentum and support and resistance levels, can be used to forecast short-term price movements (Murphy, 1999). This notion is disputed by the Efficient Market Hypothesis, which states that all the available information is already incorporated in the current price and therefore systematic predictions are unlikely (Fama, 1970; Malkiel, 2003). This study tests the hypothesis, as a null result, that any predictive model has to beat.

The case study is based on the Oracle Corporation (ORCL) due to three reasons. First, it has a 40-year continuous price record on a daily basis with different market cycles. Second, it has a strong set of pre-computed technical indicators. Third, it has a substantially growing price history, which makes stationarity a major concern for model design (*Oracle Corporation (ORCL) Stock Historical Prices & Data - Yahoo Finance*, 2026).

The paper contributes in three ways: (i) a comparison of five model families against a common naive baseline, accounting for metrics that consider class imbalance; (ii) a demonstration that an observable anecdotal pattern can be disproved and a true one validated; and (iii) two tools that translate model outputs into quantified and risk-adjusted price levels.

The study addresses four specific questions: (i) can a supervised classifier predict a 3-class Buy/Hold/Sell signal over 21 trading days with significant accuracy; (ii) can a sequence model predict the next day's closing price better than a naive baseline; (iii) can technical structure and model output be combined to establish risk-adjusted entry and exit levels; and (iv) does a correlation with lunar phases persist across different market conditions? Since this study is focused on decision support, transaction costs, slippage and position sizing are not modelled; therefore, for any claims of predictive value to be valid, each classifier must outperform the naive baseline.

2. Related Work

The trend-following and support/resistance tools utilized in this study are organized according to Murphy (Murphy, 1999). Momentum oscillators have a different origin, with Wilder (1978) introducing the Relative Strength Index and Average True Range, the latter of which is used in Section 4.7 for stop-loss sizing. Blau (1991) later introduced the True Strength Index. In contrast, Fama's (1970) review of market efficiency, along with Malkiel's (2003) revisit, argues against the presence of exploitable predictive content.

There is empirical machine learning literature that connects these perspectives (Kofi Nti, 2019), reviewed 122 studies on the application of machine learning to stock market predictions and found a common but weak statistical edge over naive models after controlling for lookahead bias. Fischer and Krauss (2018) reported a mean daily return of 0.46% and a Sharpe ratio of 5.8 (pre-cost) for an LSTM ensemble trained on the S&P 500. This result serves as a benchmark in Section 5 for comparing it with the single-ticker design of this study.

Bouman and Jacobsen (2002) documented the "Sell in May" anomaly, while Dichev and Janes (2001) observed that U.S. equity returns during new moons are roughly twice those during full moons. This finding justifies the tests on lunar phase flags discussed in Section 4.4. The risk management for stop-loss and reward-to-risk sizing is based on Elder (1993) and Van Tharp (2007). Additionally, Hyndman and Athanasopoulos recommend using a naive baseline benchmark for forecasting models.

3. Dataset

The dataset are daily OHLC bars for ORCL taken from TradingView's consolidated BATS/NYSE feed from March 12, 1986, to August 21, 2026 (TradingView, 2026). There are 10,185 sessions and 18 fields. The data can be verified with Nasdaq and Yahoo Finance (*Oracle Corporation (ORCL) Stock Historical Prices & Data - Yahoo Finance*, 2026). It includes fields such as open, high, low, and close prices; six Simple and Volume-Weighted Moving Averages (SMA and VWMA) for 21, 63, 125, 252, 504, and 756 days; and the Relative Strength Index (RSI), True Strength Index (TSI), and binary flags for New Moon and Full Moon cycles. For example, there is no field for volume.

Preprocessing included re-parsing delimiters, normalisation of UTC timestamps, and confirmation of no duplicates. A missing-value analysis revealed that the OHLC fields were complete, while the moving averages and oscillators had expected leading nulls, given their window lengths. Nulls were handled with the **dropna** method on some feature sets to allow the full date range for relevant analyses.

An outlier screening with a 3xIQR fence identified 159 of 10,184 sessions (1.56%) with extreme daily price movements, mostly in 2025 and 2026. These were kept, as they are true price changes and not data entry errors, which were re-examined in Section 4.4.

4. Methodology and Results

4.1 Stationarity, Feature Engineering, and Label Construction

The Augmented Dickey-Fuller test indicates that the closing-price series is non-stationary (statistic = −0.992, $p = 0.756$), while the daily return series is stationary (statistic = −28.492, $p < 0.0001$), supporting the modeling of returns instead of price levels (Hyndman & Athanasopoulos, 2021).

The feature set consists of daily and log returns, 20-day rolling volatility, and the percentage distance from the 21-day, 63-day, and 252-day moving averages, along with the pre-computed RSI and TSI oscillators. The classification target is a 21-trading-day forward return: "Buy" for returns above +5%, "Sell" for returns below −5%, and "Hold" otherwise. The ±5% threshold was chosen to enhance class balance (4,121 Hold, 3,573 Buy, 2,219 Sell from 9,913 rows).

Train/test partitioning is chronological (training: 1987–2018 with 7,930 rows; testing: 2018–2026 with 1,983 rows) to avoid look-ahead leakage of future information into model training.

4.2 Algorithm Selection and Justification

Table 1 justifies each model against the research questions in Section 1 and the syllabus outcome it addresses, including why each was preferred over a plausible alternative.

Table 1. Algorithm selection and justification.

Algorithm	Addresses	Why chosen over the alternative
Dummy (majority-class)	Mandatory floor	No alternative needed; defines the null every classifier must clear
Logistic Regression	RQ1 (classification)	Linear, interpretable baseline; coefficients directly expose which indicator drives each class, unlike a black-box alternative
Feed-forward Neural Network (Dense(32)→Dropout(0.2)→Dense(16)→Dense(3, softmax))	RQ1	Tests non-linear indicator interactions (e.g. RSI × volatility) that Logistic Regression's linear decision boundary cannot capture
Random Forest (GridSearchCV over TimeSeriesSplit)	RQ1	Chosen over a single Decision Tree for variance reduction through bagging, and over Gradient Boosting to reduce overfitting risks with this small (8-column) and collinear feature set.
PCA + K-Means	Regime structure	PCA is needed before clustering due to collinearity among the six SMA columns, which are transformed from the same close series. K-Means is preferred over DBSCAN because the number of regimes (bull-intensity bands) is a fixed hyperparameter rather than something to discover.
Gaussian HMM	Regime structure (probabilistic alternative)	Unlike K-Means, an HMM models the *transition* probability between regimes, not just their static membership — used as a cross-check on the K-Means result

LSTM, then Bidirectional LSTM v2 (returns-based)	RQ2 (sequence forecasting)	An LSTM is required over a flat feature table to preserve temporal ordering; the Bidirectional wrapper is safe here since it only reads within a fixed historical window, never across the prediction boundary
Rule-based ATR/support-resistance engine + empirical quantile forecast	RQ3 (decision support)	Deliberately not learned: auditability of every output level was prioritised over the marginal accuracy a learned ranking might add, given the tool is intended as a transparent decision aid

The feed-forward Neural Network was trained for up to 50 epochs (batch size 32, 15% validation split) using EarlyStopping (patience 8) and ReduceLROnPlateau (patience 4) to prevent overfitting. The LSTM variants were trained for up to 100 epochs (batch size 32, 10% validation split) with similar callbacks (patience 10 and 5, respectively). The Random Forest hyperparameter grid searched n_estimators $\in$ {100, 200}, max_depth $\in$ {3, 5, 8}, and min_samples_leaf $\in$ {5, 10, 20}, using 5-fold TimeSeriesSplit cross-validation and balanced accuracy. The chosen configuration was max_depth=3, min_samples_leaf=5, n_estimators=100.

4.3 RSI as a Statistical Signal

A Welch's t-test compares 21-day forward returns for RSI < 30 (oversold: mean −2.91%, $n = 45$) against RSI > 70 (overbought: mean +6.92%, $n = 161$), yielding a significant result ($t = -3.410$, $p = 0.0011$). This indicates a momentum-continuation effect, where oversold conditions lead to further declines and overbought conditions lead to further gains, contrary to the mean-reversion idea of traditional RSI usage (Fernando, 2026). A placebo test on the Full Moon flag shows no significant effect ($t = 0.360$, $p = 0.719$), confirming the validity of the testing procedure.

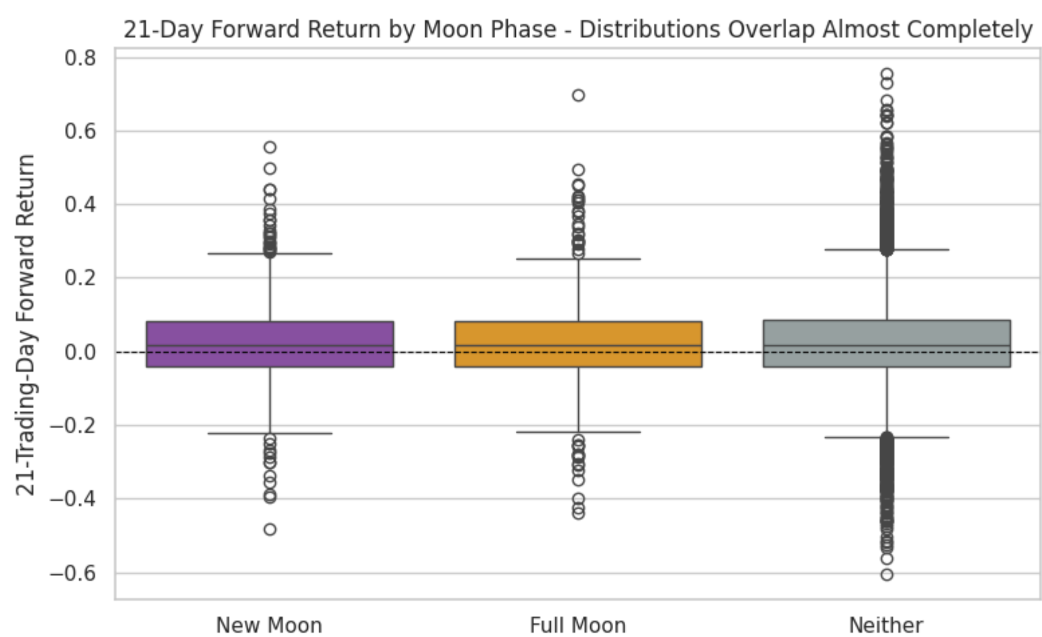
4.4 Lunar-Phase Correlation Test

A specific correlation was noted in the data: a Full Moon session on July 29, 2026, led to a rally from an intraday low of $116.20 to a high of $159.26, a gain of 37.1%. Conversely, a New Moon session on August 13, 2026, resulted in a decline to $137.43, a drop of 13.7%. These figures were verified against the source data.

Across roughly 500 New Moon and 500 Full Moon sessions from 1986 to 2026 (see Fig. 8), neither phase showed a statistically significant difference in returns at 10-day or 21-day horizons. The mean return for New Moon sessions was +2.66%, versus +2.38% for others ($t = 0.476$, $p = 0.634$), while Full Moon sessions averaged +2.61% ($t = 0.383$, $p = 0.702$). Pearson correlations for both phases remained near zero.

The frequency of volatility is a confounding factor: 3.26% of 11-trading-day windows since January 2025 had swings over 30%, compared to just 1.51% historically—about 2.2 times more frequent. Thus, the lunar-phase correlation with high-volatility sessions likely stems from this elevated regime rather than a direct lunar effect (Dichev & Janes, 2001).

Figure 8: 21-day forward return by moon phase, boxplot (New Moon / Full Moon / Neither), from notebook Section 6.7b.



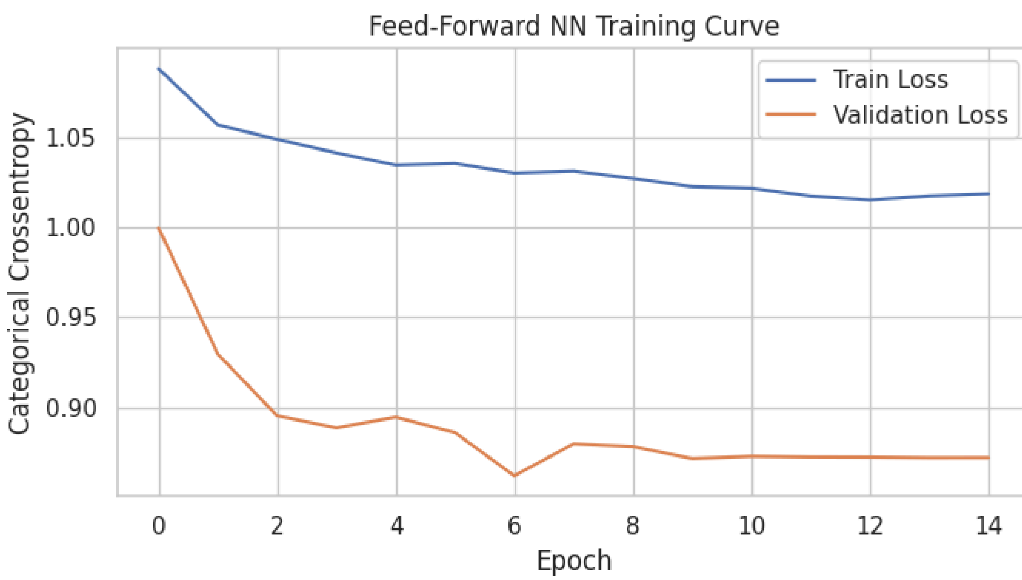
4.5 Classification Results and Generalisation Gap

Table 2. Training vs. test performance, all classifiers, 21-day horizon.

Model	Train Acc.	Train Bal. Acc.	Test Acc.	Test Bal. Acc.	Test Macro F1	Train–Test Gap (Bal. Acc.)
Baseline (majority)	41.1%	33.3%	43.5%	33.3%	20.2%	0.0 pts
Logistic Regression	47.5%	44.9%	46.2%	40.4%	39.0%	4.5 pts
Random Forest	51.4%	48.4%	45.5%	**40.5%**	**39.8%**	7.9 pts
Neural Network	52.5%[1]	46.2%[1]	46.4%	38.1%	33.3%	8.1 pts[1]

[1] Single seeded run; scikit-learn rows are exactly reproducible, this one is not bit-for-bit but falls within 1 point of repeated runs.

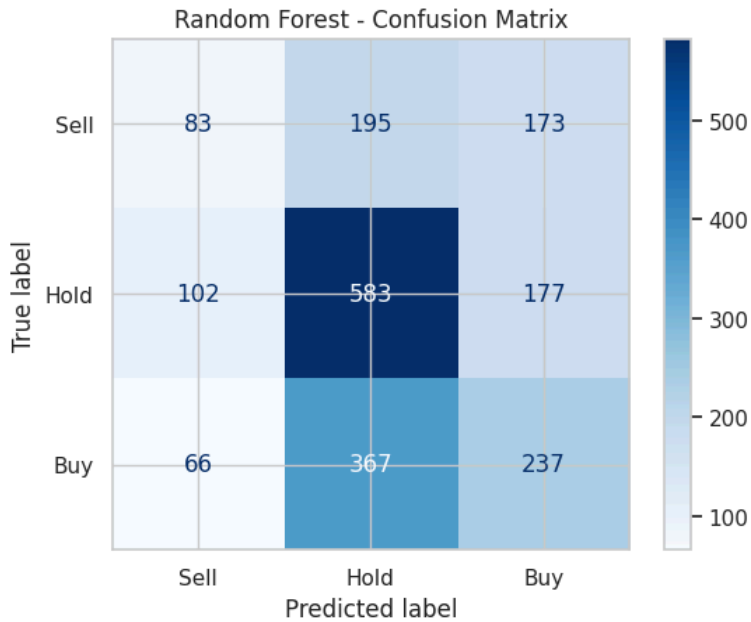
Figure 9: Neural Network training curve (train vs. validation loss), from notebook Section 9.3



[1] Neural Network figures are from a single training run, and while scikit-learn models are deterministic, exact reproduction with TensorFlow is not guaranteed. Generalisation gaps reveal model performance: the Baseline shows no gap, Logistic Regression has the smallest gap (4.5 points), suggesting limited memorisation of noise. Random Forest and the Neural Network show larger gaps (~8 points), indicating some overfitting despite regularisation techniques. However,

Random Forest achieves the highest test balanced accuracy, while the Neural Network, despite the highest raw test accuracy (46.4%), suffers from poor balanced accuracy and F1 due to low recall in the Sell class.

Figure 10: confusion matrix for the best classifier (Random Forest), from notebook Section 9.5.



In Random Forest, the most important predictor is 20-day volatility (44% importance), followed by distance from the 21-day SMA (14%). PCA condenses eight collinear features into two components explaining 97.9% of variance, while K-Means and Gaussian HMM identify net-positive-return regimes based on bullish intensity rather than direction.

4.6 Sequence Forecasting: LSTM

A price-level LSTM (Bidirectional(64) → LSTM(32) → Dense(16) → Dense(1)) is significantly outperformed by a naive persistence baseline, with an RMSE of $50.42 compared to $3.84 (see Figs. 14–15). This can be attributed to the non-stationarity discussed in Section 4.1. The training period (1986–2018) is primarily characterized by prices below $50, while the test period (2018–2026) sees prices reaching $328. This situation forces the model to extrapolate beyond the distribution of the training data.

Figure 14: LSTM v1 training curve, from notebook Section 9.8.

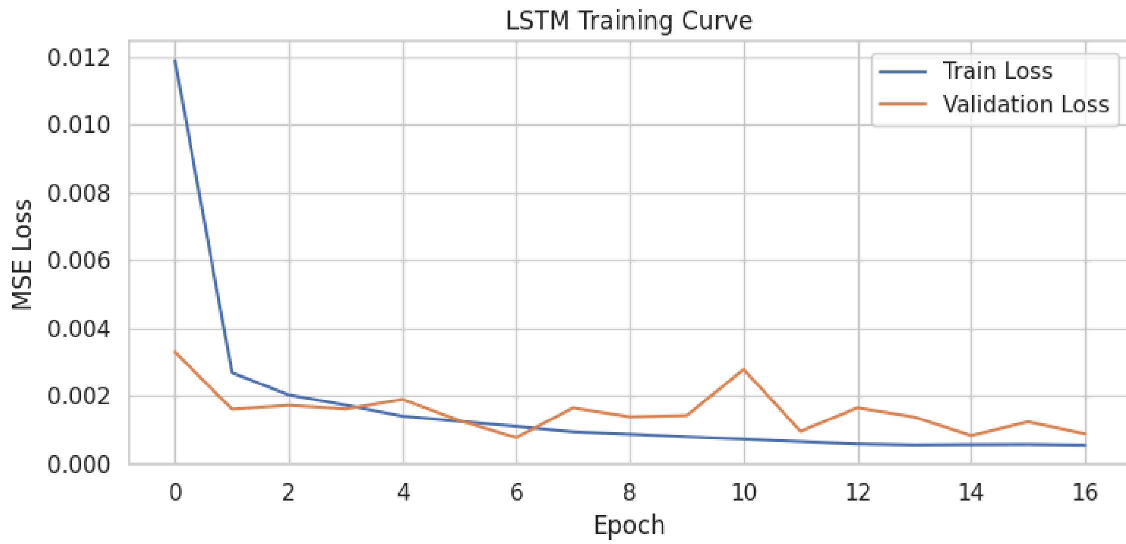
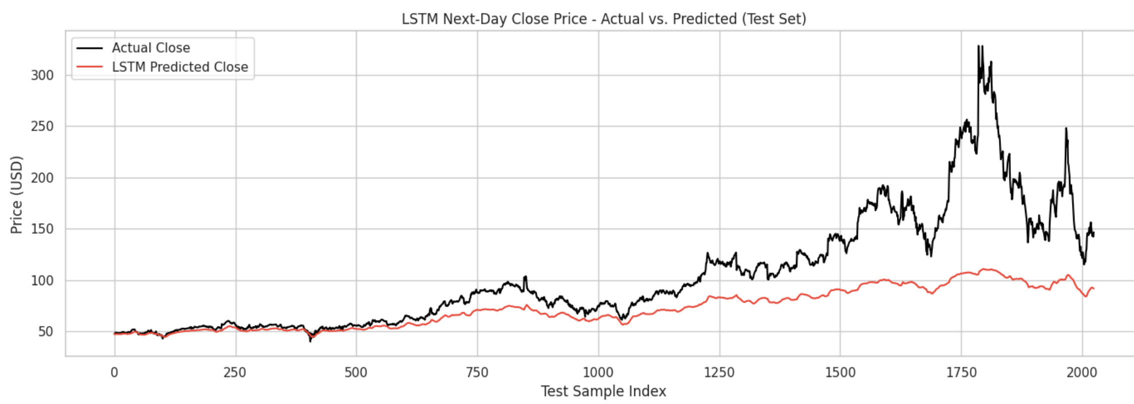


Figure 15: LSTM v1 actual vs. predicted close price, from notebook Section 9.8.



When we reformulate the same architecture and training regimen to predict daily returns instead of price levels, the performance gap nearly closes (Figs. 16–17). The RMSE is \$3.84 and the MAE is \$1.83, compared to an identical naive baseline of \$3.84. This performance is statistically indistinguishable from, and slightly better than, the persistence model. This outcome aligns with the expectations for a near-random-walk daily return series (Hyndman & Athanasopoulos, 2021) and underscores that the architecture alone does not determine forecast validity. Instead, the distinction between predicting levels and returns is the critical factor at play, reflecting the stationarity arguments outlined in Section 4.1.

Figure 16: LSTM v2 (returns-based) training curve, from notebook Section 9.8b.

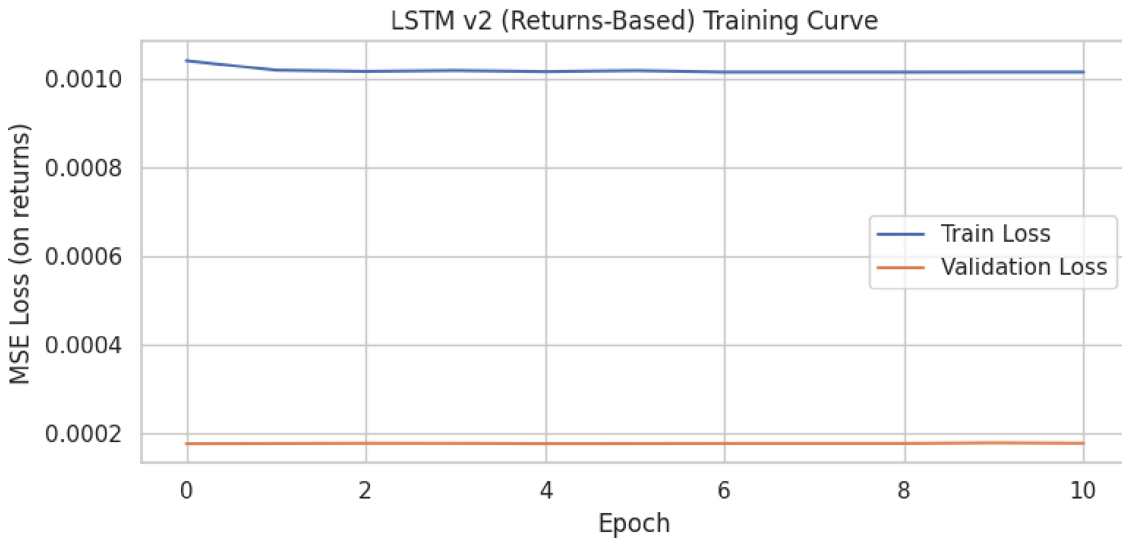
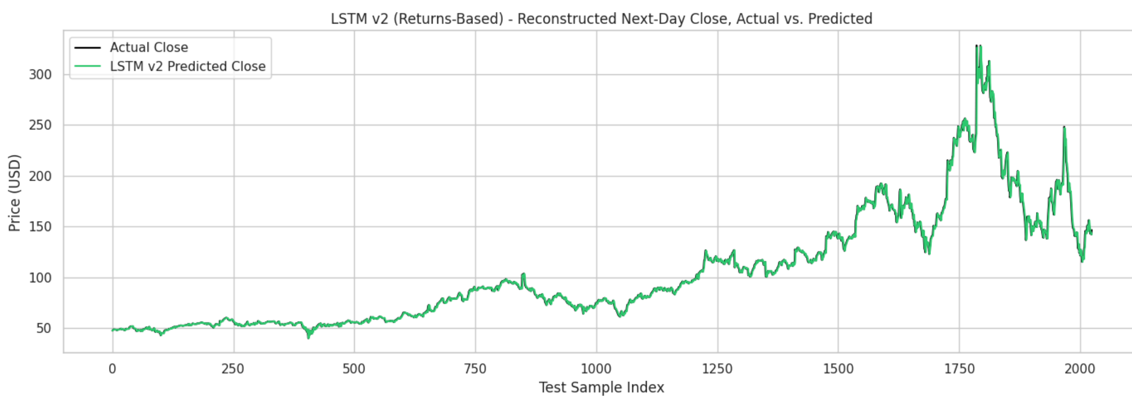


Figure 17: LSTM v2 reconstructed actual vs. predicted close price, from notebook Section 9.8b.



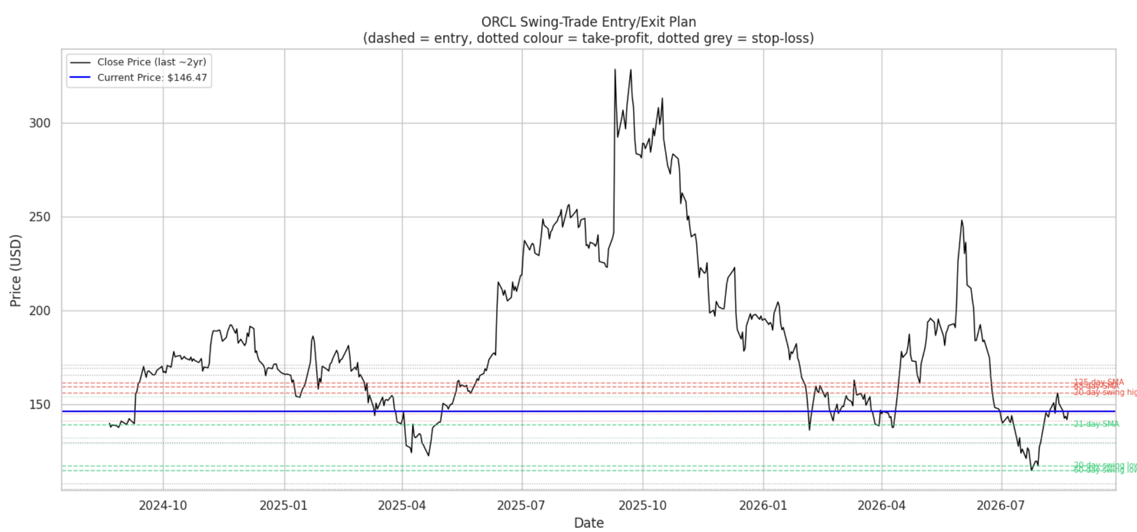
4.7 Decision-Support Outputs

Two independent forecasting methods have been developed. The first is a probability-banded 21-day forecast (Fig. 18) based on the empirical distribution of ORCL's historical 21-day forward returns. This forecast yields six price ranges anchored to the current closing price of $146.47. It includes three downside bands: mild ($134.82–$142.51), moderate ($122.06–$134.82), and severe (below $122.06), as well as three upside bands: mild ($156.45–$165.40), moderate ($165.40–$181.62), and strong (above $181.62). Each band is annotated with its historical occurrence frequency, ranging from 5% to 15%, rather than being presented as a deterministic point estimate.

Figure 18: 21-trading-day price forecast cone (6 zones), from notebook Section 10.1.

The second method is an independent technical cross-check (refer to Fig. 19) that identifies nearest support and resistance levels. It considers the 21-day Simple Moving Average (SMA) at $139.56 and the 20- and 60-day swing lows at $117.74 and $114.99 as Buy candidates. Conversely, the 20-day swing high and the 63- and 125-day SMAs at $156.22, $159.82, and $161.67 are identified as Sell candidates. Each of these is priced with a 14-day Average True Range (ATR)-derived Stop-Loss (Wilder, 1978) and features a 1.5:1 reward-to-risk Take-Profit ratio.

Figure 19: nearest support/resistance entry/exit plan, from notebook Section 10.2.



As of the most recent fully labeled session on July 23, 2026, the two forecasting methods produced different signals: the Random Forest model indicated a Sell, while the Logistic Regression model suggested a Buy. This disagreement reflects a materially different and lower-confidence state compared to model confluence, which is noted without being resolved through selection.

5. Discussion

The classification results align with the modest but statistically detectable advantage noted by Nti, Adekoya, and Weyori (2020), which is typical of rigorously validated technical machine learning (ML) studies. However, these results fall short of the cross-sectional long short-term memory (LSTM) performance reported by Fischer and Krauss (2018). This discrepancy is expected due to the single-ticker statistical power compared to a larger 500-stock panel, rather than indicating a methodological deficiency.

The generalization gap analysis presented in Table 2 offers insights not reported in comparable form by either benchmark study. Specifically, the larger train-test gap of Random Forest compared to the smaller gap of Logistic Regression highlights a trade-off between model capacity and the risk of overfitting on an 8-feature, single-asset dataset. This balance would likely play out differently in Fischer and Krauss's much larger cross-sectional sample.

Three design constraints are explicitly noted. First, a single chronological 80/20 split is used instead of walk-forward cross-validation, prioritizing scope and tractability over strict statistical rigor. Second, a classical ARIMA benchmark has been omitted, as the Augmented Dickey-Fuller (ADF) result already justifies the returns-based approach that both Random Forest and LSTM encompass while also capturing non-linear interaction effects. Lastly, the entry/exit engine (described in Section 4.7) is implemented as a transparent, rule-based layer rather than as another learned model. This choice emphasizes the auditability of every output, albeit at the cost of using static (non-adaptive) thresholds.

The lunar-phase test (Section 4.4) exemplifies the validation discipline applied throughout this study. An observed pattern is validated against the complete historical record rather than being accepted based on just two data points. This approach maintains a standard that the relative strength index (RSI) regime finding meets, while the lunar-phase claim does not. The appropriate methodological response to a striking anecdote is to verify it against the full historical record, rather than act solely based on the anecdote itself.

6. Conclusion

This study identifies a real and statistically significant technical signal in Oracle Corporation's (ORCL) price history, specifically concerning the Relative Strength Index (RSI) regime effect, with a p-value of 0.001. All three classifiers used in the analysis surpassed a random baseline in terms of balanced accuracy. The findings suggest a magnitude consistent with weak-form market efficiency, rather than strong predictability, and are directionally opposite to the conventional mean-reversion interpretation of RSI extremes.

Additionally, the generalization gap analysis indicates that the model with the best performance on the test set (Random Forest) does not fit the training data most precisely. This distinction is crucial for future model selection decisions on this dataset.

A separate test regarding the lunar-phase hypothesis, despite two verified supporting observations, does not generalize across the complete 40-year record. Instead, it can be attributed to a period of heightened volatility anticipated in 2025-2026, rather than a lunar influence. This outcome highlights the importance of analyzing anecdote-driven technical rules against a complete historical sample before applying them in practice.

The main deliverables of this study include two independent 21-day price forecasts (as detailed in Section 4.7). At the most recent evaluation point, these forecasts exhibited disagreement among classifiers rather than consensus—an outcome that is valid and actionable within an uncertainty-aware decision framework, rather than a mere null result to smooth over. Future priorities include conducting walk-forward cross-validation on all models to assess the stability of the reported generalization gaps, re-evaluating the RSI and lunar-phase effects within a narrower, more recent timeframe, and integrating a formal position-sizing framework (as outlined by Van Tharp in 2007) before any live deployment.

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Appendix A: Supplementary Figures

Fig.	Notebook Section	Description
1	6.2	Daily-return boxplot and histogram, with the IQR outlier fence
2	6.3	Interactive full-history candlestick chart w/ 6 SMAs
3	6.3	Static 5-year candlestick chart with 6 SMAs
4	6.4	Correlation heatmap of all numeric columns
5	6.4	Scatter plot, Close price vs. 21-day SMA
6	6.6	Full-history close price with support/resistance levels
7	6.6	5-year close price with support/resistance levels
11	9.6	PCA explained-variance bar chart
12	9.7	K-Means elbow plot
13	9.7	Close price coloured by detected K-Means regime

Figure 1. Daily-return boxplot and histogram, with the IQR outlier fence

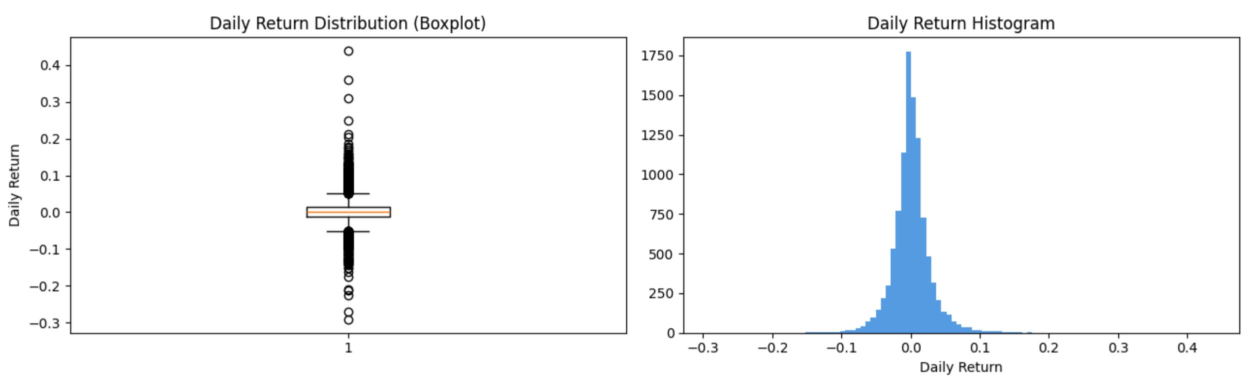

Figure 2. Interactive full-history candlestick chart w/ 6 SMAs

Figure 3. Static 5-year candlestick chart with 6 SMAs

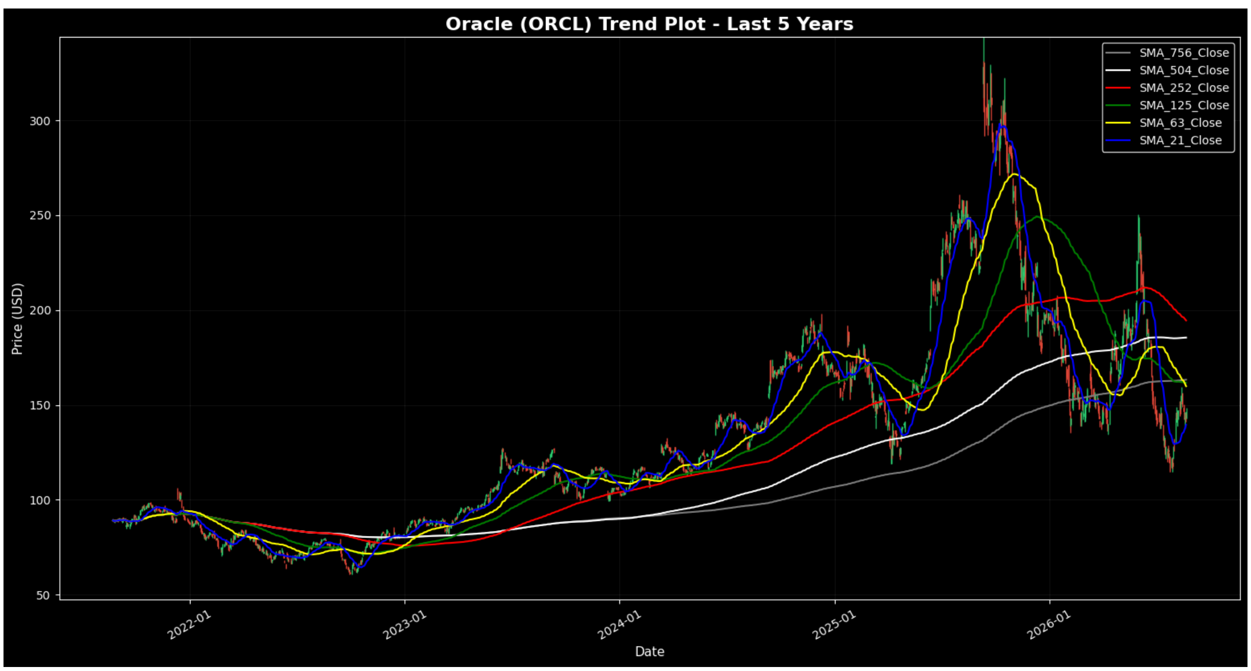

Figure 4. Correlation heatmap of all numeric columns

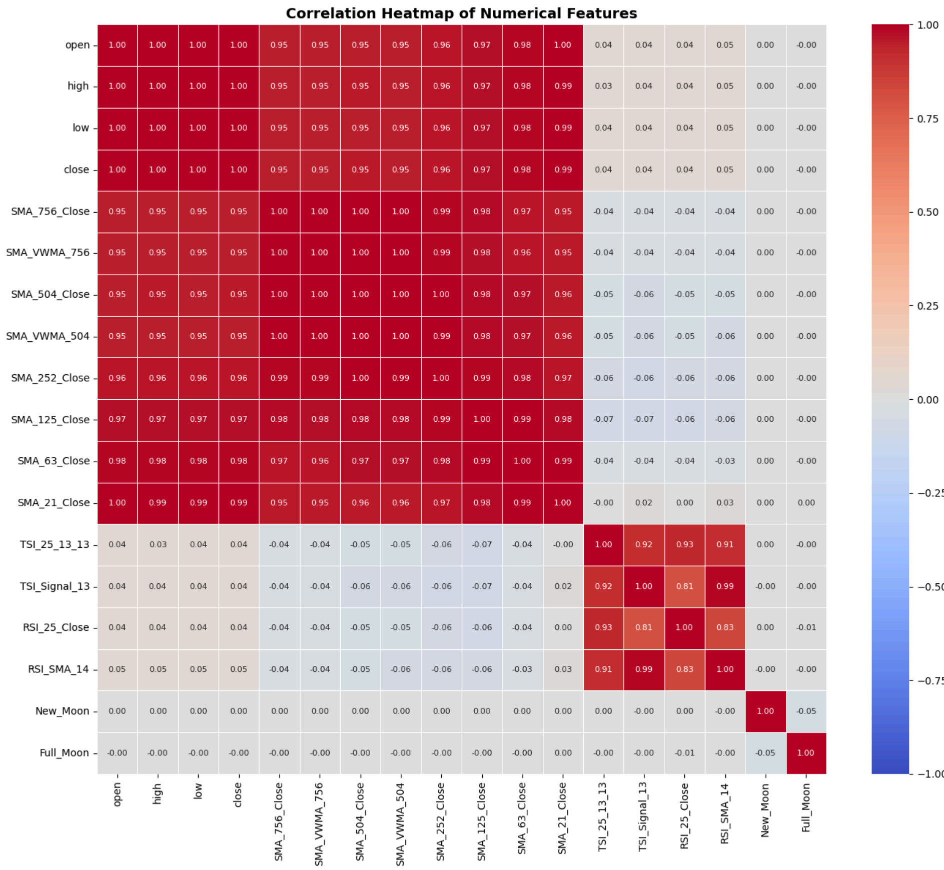

Figure 5. Scatter plot, Close price vs. 21-day SMA

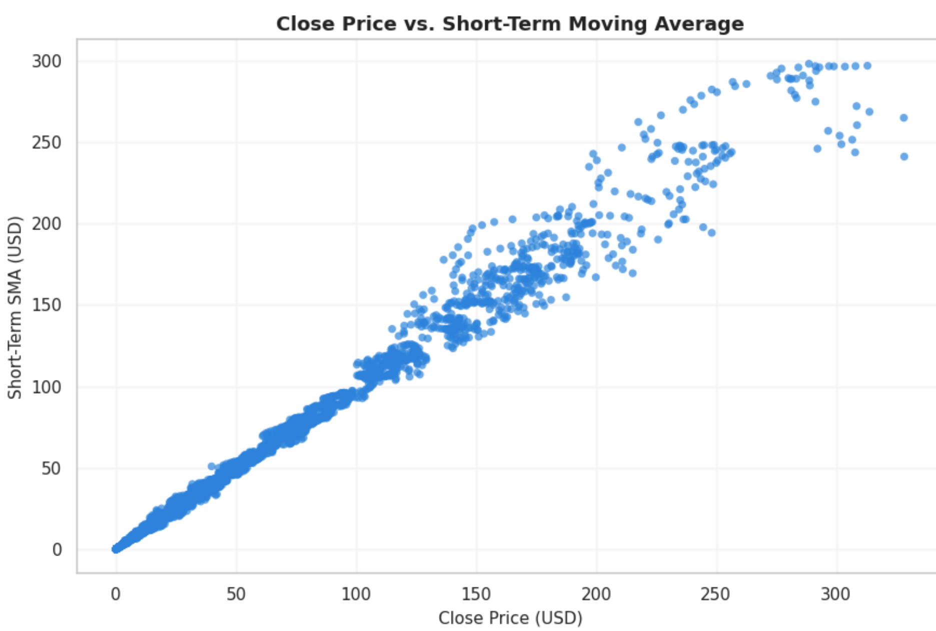

Figure 6. Full-history close price with support/resistance levels

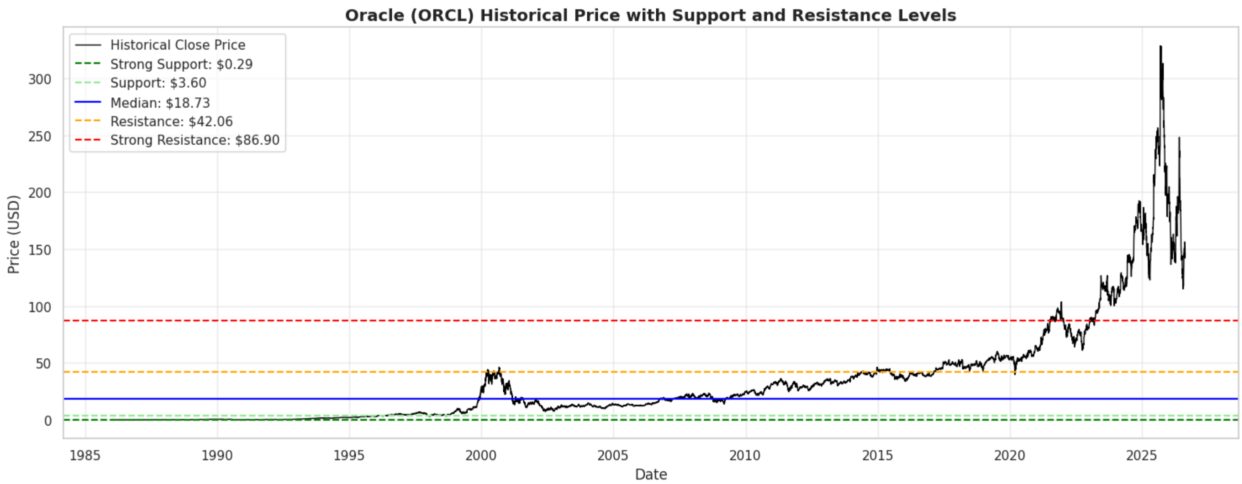

Figure 7. 5-year close price with support/resistance levels

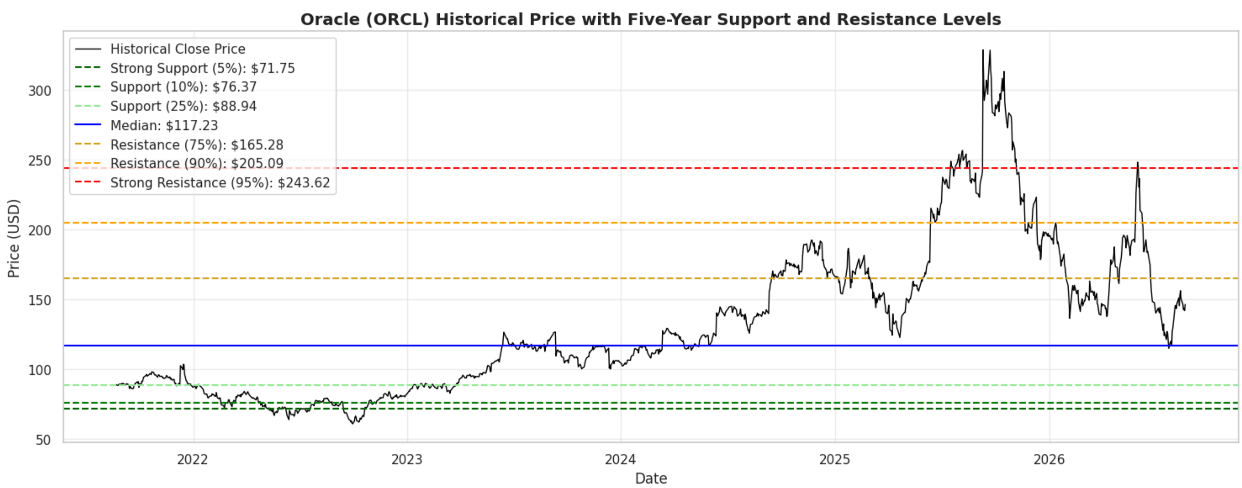

Figure 11. PCA explained-variance bar chart

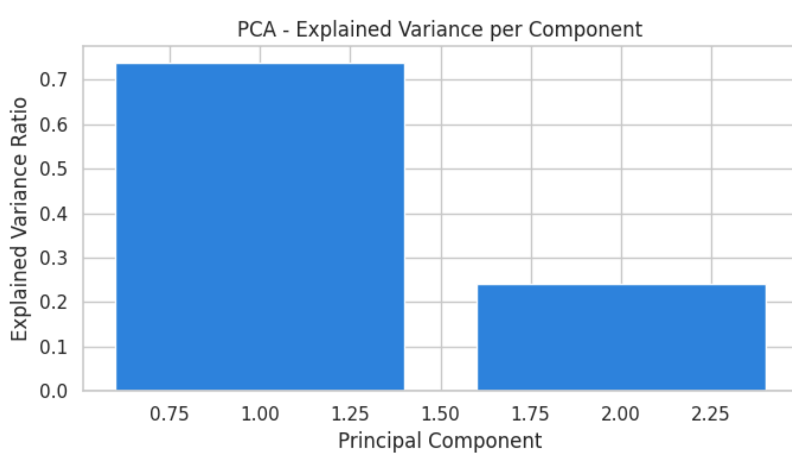

Figure 12. K-Means elbow plot

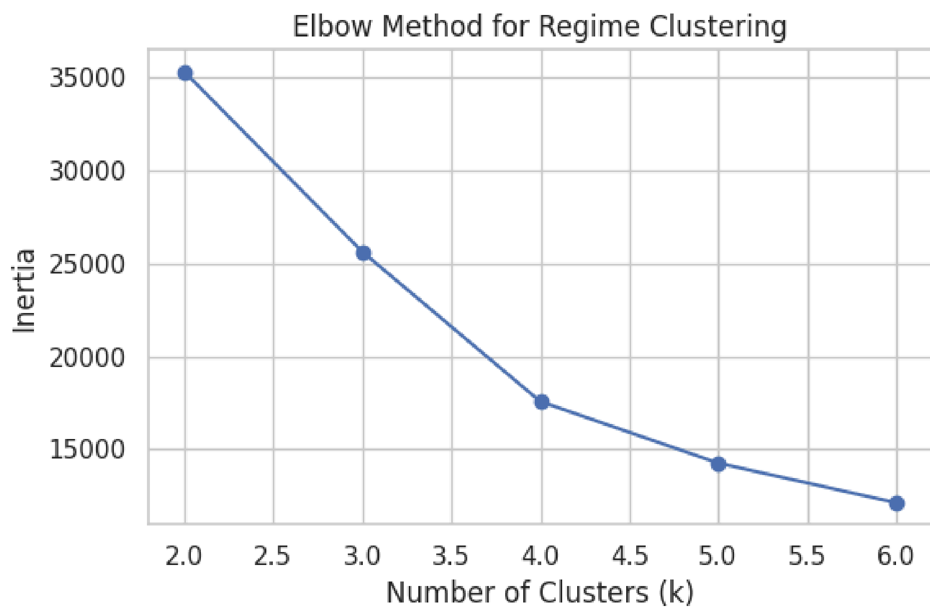

Figure 13. Close price coloured by detected K-Means regime